

Method and Proxy: The Shape of Each Tool's Question

Generated/Reviewed Documentation (Codex/Opus/Human) for v0.0.1

Contents

1	What Proxy Means Here	2
2	The Literal-to-Borrowed Spine	2
3	The Shared Kernel as Question	3
4	Bond and Ring Susceptibility Proxies	3
4.1	McConnell	3
4.2	RingSusceptibility	4
5	Aromatic Ring-Current Proxies	5
5.1	BiotSavart	5
5.2	Haigh-Mallion	5
6	Electrostatic and Multipole Proxies	6
6.1	APBSField	6
6.2	WaterField	6
6.3	EEQ	7
6.4	PiQuadrupole	7
7	Hydrogen-Bond, Contact, and Solvent Proxies	8
7.1	HBond	8
7.2	LarsenHBond	8
7.3	Dispersion	9
7.4	HydrationShell	9
8	What the Twelve Tools Make Visible	10
9	Reference Status	10

1 What Proxy Means Here

The companion on physics architecture names the hidden object: Ramsey nuclear magnetic shielding, the electronic current response to an applied field and a nuclear magnetic moment [1, 2]. The companion on categorical grounding names the empirical world in which the object is seen: amide protons report hydrogen bonds and electric fields; aromatic-near protons report ring currents; backbone carbons report peptide geometry; side-chain atoms report rotamers, contacts, and solvent exposure. This document names the shape of the tools between those two views.

“Direct expression of the physics” usually means “exact maths on a proxied source.” The maths can be exact: Biot–Savart gives the exact field of a current loop; the electric-field gradient is the exact second derivative of a point-charge potential; the dipolar kernel is the exact field shape of a point magnetic dipole. The proxy is not in the arithmetic. It is in what the arithmetic acts on. It is forced by computability: the code cannot integrate the true quantum current density of a protein frame, so it substitutes a source the maths can chew. That substitution is the proxy, baked into the technique, not bolted on afterward.

The geometric kernel is therefore the question the tool poses to the universe. Calibration is how we listen for the answer. A kernel says, for example: if this bond’s anisotropy is placed at its midpoint, is this the angular and distance pattern carried by the shielding data? If this aromatic ring is replaced by a pair of current loops, does the DFT shielding respond along that shape? If this hydrogen bond is summarized by a local table indexed by (r, θ, ρ) , does that table carry the local electronic reorganization? The proxy is the form of the question.

The recurring vocabulary is small:

The master proxy is the classical source standing in for a quantum response. A current loop, point susceptibility, point charge, quadrupole, or contact tensor carries the geometry of an effect. The coefficient that would make it a shielding contribution — $\Delta\chi$, a ring-current intensity, Buckingham’s A or γ , a quadrupole strength, C_6 , or a fitted hydrogen-bond coefficient — is the part still to be learned.

Source-shape proxies collapse extended objects to compact ones. McConnell places a distributed bond anisotropy at a midpoint. RingSusceptibility places a whole aromatic ring at a point. PiQuadrupole places an aromatic π distribution at a point axial quadrupole. What changes is not the law applied to the source; it is the shape of the source offered to the law.

Magnitude and convention proxies hold scale still so shape can be read. BiotSavart uses 1 nA. Dispersion sets $C_6 = 1$. McConnell and RingSusceptibility use unit source weights. Johnson–Bovey lobe offsets are fixed by ring type. These choices make the raw feature a measured shape, not a completed shielding.

Discretization proxies trade continuity for computable samples: polygon loops for smooth aromatic current paths, fan triangles and seven-point cubature for ring surfaces, trilinear interpolation for potential and DFT grids, and central differences for derivatives of a stored scalar potential.

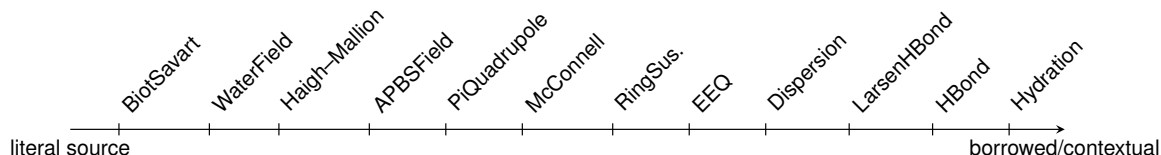
Model-substitution proxies replace an inaccessible source by a computable model of it: Poisson–Boltzmann continuum electrostatics for a solvated protein environment, fixed three-site water charges for explicit solvent electron density, EEQ charges for the real ground-state charge distribution, and Larsen donor/acceptor archives for hydrogen-bond chemistry.

These are not ranks of blame. They are names for how method reaches toward physics.

2 The Literal–to–Borrowed Spine

The spectrum that matters is how literal the source is, not how exact the arithmetic is. At the literal end, BiotSavart really computes the magnetic field of a current, although the ring current itself has been turned

into two unit-current polygon loops. Near it, WaterField really computes fields and field gradients from point charges, and Haigh–Mallion really integrates a dipolar kernel over a ring surface. In the middle, McConnell and RingSusceptibility use real dipolar angular physics after collapsing an extended anisotropy to a point. Farther along, Dispersion borrows the rank-2 angular shape but changes the distance law to r^{-6} , and HBond borrows the same McConnell tensor as a way to ask whether a chemically real hydrogen-bond axis carries shielding. At the contextual edge, HydrationShell emits solvent arrangement rather than a field or tensor. LarsenHBond is the special case: its source is already DFT shielding data, so its proxy lies in classing, interpolation, and local scope, not in a borrowed geometric kernel.



This line is not a scorecard. It is a reading order. Moving right does not make the method less useful; it changes what the method is asking. A literal current loop asks for a magnetic field. A hydrogen-bond axis asks whether a strong local chemical direction can carry a shielding response after calibration. A hydration half-shell asks whether the local solvent arrangement conditions the same response.

3 The Shared Kernel as Question

Most tools return to the Hessian of $1/\rho$,

$$D_{ab}(\rho) = \frac{3\rho_a\rho_b - \delta_{ab}\rho^2}{\rho^5} = \frac{3\hat{\rho}_a\hat{\rho}_b - \delta_{ab}}{\rho^3}.$$

ρ is the displacement from a source point to the observed atom, $\rho = |\rho|$, and δ_{ab} is one when the Cartesian directions a and b are the same. The tensor is symmetric and traceless. Along the line of sight it has eigenvalue $+2/\rho^3$; in the two perpendicular directions it has $-1/\rho^3$. The three values sum to zero.

That one tensor is both a magnetic and an electric object. In magnetostatics it is the angular field shape of a point dipole. In electrostatics it is the electric-field gradient of a point charge away from the charge. In McConnell, RingSusceptibility, and HBond it is folded into an asymmetric susceptibility kernel whose trace average is

$$f(\rho, \hat{u}) = \frac{3(\hat{\rho} \cdot \hat{u})^2 - 1}{\rho^3}.$$

$\hat{u}$ is the source axis: a bond direction, a ring normal, or an $N \cdots O$ hydrogen-bond axis. In Dispersion the same angular matrix is kept but the distance law becomes r^{-6} . In PiQuadrupole the $1/r$ source law is differentiated two steps farther. The repeated form is not accidental. It is the common grammar by which these classical sources ask about a rank-2 shielding response.

4 Bond and Ring Susceptibility Proxies

4.1 McConnell

McConnell asks: if a covalent bond is an anisotropic magnetic object, what rank-2 shape does that bond cast at a nearby nucleus? The physical reach is bond and group anisotropy, especially peptide $C = O$, peptide $C - N$, side-chain carbonyls, aromatic bonds, and other covalent axes that move when backbone and side-chain geometry move [5, 3].

The code’s source is a bond midpoint $\mathbf{m}$, a bond length ℓ , and a unit bond axis $\hat{\mathbf{b}}$. For an atom at $\mathbf{r}$, $\mathbf{x} = \mathbf{r} - \mathbf{m}$, $r = |\mathbf{x}|$, $\hat{\mathbf{d}} = \mathbf{x}/r$, and $c = \hat{\mathbf{d}} \cdot \hat{\mathbf{b}}$. The full per-bond tensor is

$$M_{ab} = \frac{9c \hat{d}_a \hat{b}_b - 3\hat{b}_a \hat{b}_b - (3\hat{d}_a \hat{d}_b - \delta_{ab})}{r^3}.$$

The scalar $f = (3c^2 - 1)/r^3$ is the trace average of this full tensor for a nondegenerate bond. The asymmetric first dyad can populate T_1 ; the symmetric traceless remainder populates T_2 .

The proxy is source-shape and magnitude at once. A bond is not a point; its electron density and susceptibility anisotropy are distributed along an extended covalent region. McConnell collapses that distribution to the midpoint and asks the dipolar law to speak from there. The calculator also assigns unit weight. It does not multiply by a bond-specific $\Delta\chi$, so the shape is computed and the material strength is left for calibration.

Its numerical guards also express the source-shape proxy. The atom path omits a target closer than half the bond length to the midpoint, because inside that ball the midpoint no longer stands cleanly for the extended bond. The singular guard at 0.1 Å, the self-source filter, and the 10 Å midpoint query decide membership in the question; they do not taper or rescale accepted terms.

On the spectrum, McConnell is middle. The angular law is physical and old. The point midpoint, unit bond weight, and missing $\Delta\chi$ are the proxy. The tool reaches toward anisotropic covalent shielding in the atom categories where local bond geometry is one of the real determinants.

4.2 RingSusceptibility

RingSusceptibility asks: if the aromatic ring were a single anisotropic susceptibility at its centre, with its principal axis along the fitted normal, what inverse-cube tensor would reach the atom? It takes the McConnell question and replaces the bond axis $\hat{\mathbf{b}}$ by the ring normal $\hat{\mathbf{n}}$. With $\mathbf{x} = \mathbf{r} - \mathbf{c}$, $r = |\mathbf{x}|$, $\hat{\mathbf{d}} = \mathbf{x}/r$, and $u = \hat{\mathbf{d}} \cdot \hat{\mathbf{n}}$, the scalar is

$$f_{\text{ring}} = \frac{3u^2 - 1}{r^3},$$

and the full tensor is the same three-term M_{ab} with $\hat{\mathbf{b}} \mapsto \hat{\mathbf{n}}$.

The proxy is a strong source-shape compression. The ring’s π system has extent, vertices, height above the plane, and a current path. This calculator keeps the centre and the normal, obtained from the shared SVD ring geometry, and asks only the far-field point-susceptibility question. It does not multiply by a ring-current intensity, a ring-type susceptibility, a Johnson–Bovey lobe offset, or a fitted scale. All accepted rings carry unit geometric weight.

The near-field exclusion is therefore meaningful. A target inside the mean ring radius is not being asked of the point-source proxy on the atom path; ring vertices and their bonded neighbours are also omitted because they belong to a through-bond aromatic regime rather than a through-space ring-source regime. The accepted tensor is decomposed raw, so the T_0 , T_1 , and T_2 channels retain the complete point-susceptibility question.

On the spectrum, RingSusceptibility sits just beyond McConnell. It uses the same literal dipolar angular law, but the source it compresses is a whole aromatic ring rather than one bond. It reaches toward the ring-current and ring-susceptibility category for aromatic-proximal nuclei, especially protons whose shifts have long required explicit aromatic terms [4, 8, 10].

5 Aromatic Ring-Current Proxies

5.1 BiotSavart

BiotSavart asks the most literal magnetic question in the suite: what secondary magnetic field would a circulating aromatic current create at the atom? For a current I flowing along a curve, the field is

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{s})}{|\mathbf{r} - \mathbf{s}|^3}.$$

$d\boldsymbol{\ell}$ is a directed element of the current path at source point $\mathbf{s}$. $\mathbf{r} - \mathbf{s}$ points from that current element to the field point. The cross product supplies the curling direction of the magnetic field.

The code does not integrate over a smooth quantum current density. It replaces the aromatic current by the Johnson–Bovey double loop: two polygonal copies of the ring, displaced by a fixed ring-type lobe offset on either side of the fitted ring plane [6]. Each lobe carries half the requested current. Each polygon edge is a straight wire segment, and the segment field is evaluated by the closed-form finite-wire Biot–Savart expression, not by quadrature.

The magnitude proxy is explicit. Atom calculations call the double-loop field with 1 nA. The code records the ring intensity as provenance but does not multiply it into the tensor. The shielding-like matrix is

$$G_{ab} = -10^6 B_a^{(1)} n_b.$$

$B_a^{(1)}$ is component a of the unit-current field, and n_b is component b of the ring normal. The column index says only the applied-field component normal to the ring drives this current in the model. The row index is the secondary-field component produced at the atom. The outer product has rank at most one and can be asymmetric.

On the spectrum, BiotSavart is closest to literal. Its arithmetic is the real field law of a current. The proxies are the current source itself: polygon loops for the unknown quantum current path, fixed lobe offsets for the two π lobes, and unit current for uncalibrated strength. It reaches toward the aromatic ring-current shielding term named by Pople, Johnson–Bovey, and later protein calibrations [4, 6, 8, 9].

5.2 Haigh–Mallion

Haigh–Mallion asks: what field-like tensor is obtained if the aromatic ring interior is read as a surface of dipolar response? The raw object is

$$H_{ab}(\mathbf{r}) = \int_S \left(\frac{3\rho_a \rho_b}{\rho^5} - \frac{\delta_{ab}}{\rho^3} \right) dS.$$

S is the fan-triangulated ring surface, dS is an area element, and $\boldsymbol{\rho} = \mathbf{r} - \mathbf{s}$ points from a surface point $\mathbf{s}$ to the atom. The integrand is the same dipolar kernel. The reported kernel contracts the surface tensor with the ring normal and forms

$$V_a = H_{ab} n_b, \quad G_{ab} = -V_a n_b.$$

The proxy is neither a current wire nor a point susceptibility. It is a surface reading of the ring response, inherited from the Haigh–Mallion lineage [7]. The ring interior is not flattened onto the fitted plane; the fan uses the actual ring centre and vertices. But continuity is traded for tractable quadrature. Each fan triangle receives a seven-point degree-five rule, with depth-two adaptive subdivision near the target. Degenerate triangles and near-singular quadrature points are omitted rather than repaired by a different source.

The magnitude proxy is again the absence of the electronic response coefficient. The surface integral carries the geometry and the normal projection; calibration must decide the scale. The raw $\mathbf{H}$ is

symmetric and traceless in exact arithmetic because it integrates symmetric traceless kernels. The reported G , after normal contraction and outer product, can populate all three $T_0/T_1/T_2$ channels.

On the spectrum, Haigh–Mallion is close but less literal than BiotSavart. It does not compute the field of a current path. It asks a surface-kernel version of the same aromatic response question, retaining ring extent more richly than RingSusceptibility and leaving strength for calibration.

6 Electrostatic and Multipole Proxies

6.1 APBSField

APBSField asks: what electric field and electric-field-gradient tensor does a continuum electrostatic model assign to each atom in this conformation? The physics bridge is Buckingham’s expansion: shielding can change with local electric fields and field gradients, with coefficients that depend on the nucleus and local electronic structure [11, 13, 14].

The source proxy is model substitution. APBS solves a linearized Poisson–Boltzmann problem for a solvated biomolecule on a grid [12]. The calculator receives only the scalar potential ϕ , in APBS units. It does not receive APBS-internal fields. At an atom $\mathbf{r}_i$, it forms a central finite difference,

$$E_a(\mathbf{r}_i) = -\frac{\phi(\mathbf{r}_i + h_a \hat{e}_a) - \phi(\mathbf{r}_i - h_a \hat{e}_a)}{2h_a}.$$

h_a is the grid spacing along axis a , and $\hat{e}_a$ is the Cartesian unit vector. The minus sign is $\mathbf{E} = -\nabla\phi$. The field-gradient entry is then another central difference,

$$G_{ab}(\mathbf{r}_i) = \frac{E_a(\mathbf{r}_i + h_b \hat{e}_b) - E_a(\mathbf{r}_i - h_b \hat{e}_b)}{2h_b}.$$

The scalar potential at each off-grid point is trilinearly interpolated from the stored grid.

The discretization proxy is therefore central to the method: a continuous continuum potential becomes a finite grid; a point atom samples a trilinear interpolant; derivatives become nested finite differences. The code then symmetrizes the computed G and removes its trace before exporting the five T_2 entries. The trace removal asks for the external rank-2 field shape rather than the grid-smeared self-potential average.

On the spectrum, APBSField is literal about electrostatic field derivatives but not literal about the molecular source. It reaches toward electric-field shielding categories for amide protons, nitrogens, carbonyls, side chains, and solvent-exposed atoms, using a continuum potential as the source the derivative maths can act on.

6.2 WaterField

WaterField asks the parallel explicit-solvent question: what field and electric-field gradient arise from the accepted water point charges? A water enters when its oxygen lies inside the configured oxygen cutoff; once admitted, all three charge sites contribute. For charge q_j at site $\mathbf{s}_j$ and target atom $\mathbf{x}_i$, the code uses

$$\mathbf{r} = \mathbf{s}_j - \mathbf{x}_i$$

and forms

$$\mathbf{E}_j = k_e q_j \frac{\mathbf{r}}{r^3}, \quad V_{ab}^{(j)} = k_e q_j \left(\frac{3r_a r_b}{r^5} - \frac{\delta_{ab}}{r^3} \right).$$

k_e is the Coulomb conversion used by the code, with charges in elementary charge units and distances in Å. The tensor is the exact Hessian shape of a point-charge potential away from the charge. After summing, the code subtracts the residual trace before writing the EFG T_2 features.

The proxy is the charge source. Real water has an electron density, molecular polarizability, and mutual polarization with the protein. WaterField replaces that by fixed topology point charges at the oxygen and hydrogens, then applies exact point-charge arithmetic to that substituted source. It also truncates the real-space sum with no Ewald tail, continuum reaction field, or induced polarization term inside the calculator.

On the spectrum, WaterField is close to literal for the mathematical object it computes. It is a direct electrostatic field and EFG from explicit point charges. It reaches toward the same Buckingham-style electric-field shielding route as APBSField, but with solvent molecules as discrete sources rather than a dielectric continuum.

6.3 EEQ

EEQ asks an upstream source question rather than a shielding question: given the geometry, what partial charge distribution equalizes the atomic charge energy under a fixed total charge? With effective electronegativities χ_i^{eff} , a symmetric Coulomb/hardness matrix A , and total charge Q , the model minimizes

$$E(\mathbf{q}) = \sum_i \chi_i^{\text{eff}} q_i + \frac{1}{2} \sum_{ij} A_{ij} q_i q_j, \quad \sum_i q_i = Q.$$

$\mathbf{q}$ is the vector of atomic charges. χ_i^{eff} is a coordination-dependent electronegativity. A_{ij} contains the local hardness and regularized Coulomb coupling. The code factors the positive block A by Cholesky, solves two right-hand sides, and recovers the Lagrange multiplier enforcing the charge sum.

The proxy is model substitution for the electron density. EEQ does not pretend to compute the shielding tensor. It replaces the real many-electron charge distribution by scalar atom charges determined by smooth coordination counts, element parameters, and an Ohno–Klopman interaction. It writes charges and coordination numbers, not fields. Those charges can be viewed architecturally as possible sources for field and field-gradient kernels. The D4/EEQ lineage supports this kind of geometry-dependent charge model [15, 16].

On the spectrum, EEQ sits off the direct kernel line. It is more literal than a pure descriptor because it solves a constrained electrostatic model, but its output is a source-making proxy, not a shielding proxy. It reaches toward the electrostatic categories by supplying a computable charge distribution that the true electronic density cannot provide cheaply.

6.4 PiQuadrupole

PiQuadrupole asks: if the aromatic π distribution is read as a point axial quadrupole at the ring centre, what electric-field-gradient shape reaches the atom? This is a multipole question. Multipoles are the language by which a localized charge distribution is described from far away [17].

For atom displacement $\mathbf{d} = \mathbf{r}_i - \mathbf{c}$, distance $r = |\mathbf{d}|$, and height $h = \mathbf{d} \cdot \hat{\mathbf{n}}$, the code evaluates the contraction

$$G_{ab} = T_{abcd} \hat{n}_c \hat{n}_d, \quad T_{abcd} = \partial_a \partial_b \partial_c \partial_d \frac{1}{r}.$$

$\hat{\mathbf{n}}$ is the fitted ring normal. T_{abcd} is the fourth derivative of the Coulomb source law. The source-side two derivatives belong to the quadrupole; the target-side two derivatives make the field gradient. After algebraic contraction, every tensor entry has units $\text{\AA}^{-5}$. The calculator also writes a scalar

$$q_{\text{quad}} = \frac{3(h/r)^2 - 1}{r^4},$$

which is a separate angular field kernel, not the trace of G .

The proxy is source-shape and magnitude. The aromatic π cloud is not a point quadrupole; the method places an axial quadrupole at the centre and uses the normal as its axis. It multiplies neither by a physical quadrupole strength nor by a Buckingham shielding-response coefficient. The near-field exclusion at the mean ring radius marks where a point quadrupole is no longer the question being asked of an atom inside the source’s own size.

On the spectrum, PiQuadrupole is middle-literal. Multipole expansion is a literal far-field language for charge distributions, and the fourth derivative is exact for the point axial source. The proxy is the point axial source and unit strength. It reaches toward aromatic electrostatic and multipolar effects that can couple to shielding through electric-field-gradient response [11, 18].

7 Hydrogen-Bond, Contact, and Solvent Proxies

7.1 HBond

HBond asks: if a DSSP-resolved backbone hydrogen bond supplies a strong local axis, does the shielding data carry an inverse-cube rank-2 response around that axis? DSSP supplies donor and acceptor residue partners by the Kabsch–Sander hydrogen-bond criterion [19]. The calculator resolves each accepted pair to donor backbone N and acceptor backbone O , then reduces the segment to midpoint m , length L , and unit direction $\hat{h}$ from N to O .

For target atom r_i , the tensor is exactly the McConnell family with $\hat{b} \mapsto \hat{h}$:

$$M_{ab} = \frac{9c \hat{d}_a \hat{h}_b - 3\hat{h}_a \hat{h}_b - (3\hat{d}_a \hat{d}_b - \delta_{ab})}{r^3}, \quad c = \hat{d} \cdot \hat{h}.$$

$\hat{d}$ points from the hydrogen-bond midpoint to the target atom, and r is that midpoint-to-atom distance. The scalar sum is $(3c^2 - 1)/r^3$. The raw asymmetric sum is decomposed into the full $T_0/T_1/T_2$ packet.

The proxy is deliberately borrowed. A hydrogen bond is not a magnetic susceptibility rod. The real physical event is hydrogen-bond electronic reorganization: donor and acceptor electron densities polarize, and the shielding response of amide protons, nitrogens, and nearby carbonyls changes [20, 21]. HBond keeps the chemically meaningful local axis and borrows the McConnell tensor as the question shape. It does not multiply by DSSP energy, donor angle, acceptor chemistry, charge, or an empirical hydrogen-bond shielding coefficient.

On the spectrum, HBond is far and useful. Its distance law and tensor shape are not a literal hydrogen-bond field. They are a geometric form that says: there is a strong local axis here; calibration can tell us whether shielding listens to it.

7.2 LarsenHBond

LarsenHBond asks a different hydrogen-bond question: given a local donor–acceptor geometry, what tensor-valued DFT shielding term is stored in the ProCS/Larsen parameterization? Its source is not a borrowed dipolar law. The grid entries are DFT shielding-difference tensors in ppm [21].

The lookup coordinates are

$$r = \|\mathbf{o} - \mathbf{h}\|, \quad \theta = \angle(\text{H} - \text{O} - \text{C}), \quad \rho = \text{dihedral}(\text{H} - \text{O} - \text{C} - \text{third}).$$

$\mathbf{h}$ is the donor hydrogen position. $\mathbf{o}$ is the acceptor oxygen. C is the heavy atom bonded to that oxygen, and third fixes the acceptor-side plane. Donor class and acceptor class select one of six archives. The

(r, θ, ρ) point selects a cell, with periodic wrapping in ρ , and each tensor entry is trilinearly interpolated from the eight surrounding grid corners. The canonical donor-frame tensor is rotated into the protein frame by

$$\sigma_{\text{lab}} = R^T \sigma_{\text{canon}} R.$$

Both tensor indices rotate because shielding is rank two.

Here the proxy is interpolation and classing. A continuous landscape of hydrogen-bond electronic structure is represented by six donor/acceptor archives and a finite tensor grid. Side-chain carbonyl acceptors fold into the backbone-carbonyl archive. The runtime interpolation is trilinear even though the dense grids carry upstream smoothing. Guarded donor-frame degeneracies fall back to the identity frame. The water offset is a scalar amide-H term, not part of the tensor sum.

On the spectrum, LarsenHBond is close to electronic structure in source but local in scope. It is not a complete chemical shift; it is a local empirical shielding contribution. Its proxy is not a geometric stand-in for a hydrogen-bond field. It is the discretized, classed, rotated form by which DFT hydrogen-bond shielding data can be asked of a protein conformation.

7.3 Dispersion

Dispersion asks: where do short-range aromatic contacts present a polarizable r^{-6} environment to the target atom? London dispersion is a real electronic interaction from correlated fluctuations, with pair strength carried by C_6 [22, 23, 24]. The calculator keeps the distance law and an orientation-resolved rank-2 shape, while setting the coefficient to one.

For a target atom at r_i , an aromatic vertex at v_j , and $d = r_i - v_j$, the per-vertex tensor is

$$K_{ab} = \frac{S(r)}{r^6} \left(3\hat{d}_a\hat{d}_b - \delta_{ab} \right), \quad r = |d|.$$

$S(r)$ is the Brooks–CHARMM switching function: one through the switching onset, smoothly tapered, and zero at the cutoff. The scalar contact stored beside the tensor is $S(r)/r^6$, not the trace of K , because the tensor is traceless in exact arithmetic.

The proxy is mixed. The r^{-6} distance law is the dispersion signature. The angular tensor is a borrowed rank-2 descriptor, not a Maxwell field from a permanent dipole. The source is restricted to aromatic vertices, not all atom-pair polarizabilities, and $C_6 = 1$ holds the chemical coefficient for calibration. The switching window is a discretization-in-time kindness to the trajectory feature: the contact vanishes with continuous value and slope as a vertex leaves the cutoff.

On the spectrum, Dispersion sits between model and metaphor. It reaches toward real polarizable contact effects, including shielding changes from van der Waals interactions [25], but its emitted tensor is an orientation-resolved contact question rather than a completed dispersion shielding formula.

7.4 HydrationShell

HydrationShell asks a contextual question: what local solvent arrangement does each atom live inside? It does not emit a tensor, a field derivative, or a chemical shift. It emits first-shell water geometry, water dipole orientation, nearest ion distance and charge, and, through the sibling hydration geometry path, SASA-normal water polarization. This belongs beside shielding because protein electronic response is conditioned by water and the dielectric boundary structure [26, 27, 28].

For the centroid half-shell, the code forms the coordinate centroid

$$c = \frac{1}{N} \sum_{j=1}^N r_j$$

and the atom-to-interior direction

$$\hat{\mathbf{u}}_i = \frac{\mathbf{c} - \mathbf{r}_i}{|\mathbf{c} - \mathbf{r}_i|}.$$

Water oxygens within the first-shell cutoff are split by the sign of $\hat{\mathbf{r}}_{iw} \cdot \hat{\mathbf{u}}_i$. The output

$$h_i = \frac{n_{\text{exposed},i}}{n_{\text{exposed},i} + n_{\text{buried},i}}$$

is an exposed-side fraction when the denominator is nonzero. The water dipole direction uses

$$\boldsymbol{\mu}_w = q_H(\mathbf{h}_{1w} - \mathbf{o}_w) + q_H(\mathbf{h}_{2w} - \mathbf{o}_w),$$

then averages the cosine between $\boldsymbol{\mu}_w$ and the atom-to-water-oxygen direction for waters with nonzero dipole magnitude.

The proxy is descriptive, not field-theoretic. A first-shell oxygen cutoff stands in for a structured hydration layer. A centroid or SASA normal stands in for inside/outside direction. A dipole cosine stands in for solvent orientation around the atom. A nearest-ion scalar stands in for ionic context. None of these numbers claims to be σ , D_{ab} , or a Buckingham field gradient. They ask whether the solvent state around a nucleus helps explain variation in the electronic response measured elsewhere.

On the spectrum, HydrationShell sits at the contextual edge. It is far from a literal shielding kernel because its question is not “what field is produced?” but “what local solvent situation is present?” That is still a physical question. It is the environmental frame in which the more direct kernels act.

8 What the Twelve Tools Make Visible

The suite is not twelve approximations to the same formula. It is twelve ways of making a computable question out of one inaccessible quantum response.

BiotSavart keeps the current-field mechanism and proxies the current source. Haigh–Mallion keeps ring extent through a surface integral and proxies the surface response. RingSusceptibility collapses that same aromatic category to a point anisotropy. McConnell collapses bond anisotropy to a midpoint. APBSField and WaterField both ask for electric field gradients, one from a continuum potential grid and one from explicit point charges. EEQ asks what charge source a geometry-dependent equilibration model would provide. PiQuadrupole asks the higher-multipole version of the electric-field-gradient question. HBond borrows an inverse-cube tensor to test a chemically real axis. LarsenHBond carries DFT hydrogen-bond tensors through classed interpolation and frame rotation. Dispersion gives a short-range polarizable contact a rank-2 angular form. HydrationShell records solvent context rather than a tensor.

The architecture companion says these are shadows of one quantum object. The categorical companion says why different nuclei and geometries made the field name different measures. The method-and-proxy view adds the missing middle: each calculator is the concrete shape of a question, exact where exact calculation is available, proxied where the source must become computable, and open to calibration where the electronic response coefficient has not yet been heard.

9 Reference Status

The bibliography below reuses the verified companion-chapter set. In this pass, the DOI subset cited here was checked with Crossref using the requested `nmr-refcheck/0.1` user agent. Crossref returned DOI metadata for every cited DOI. Europe PMC `resultType=core` records exposed abstract text for the modern support papers used for ab initio shift determinants, ring-current calibration, electric fields, EEQ/D4, multipoles, hydrogen bonding, dispersion, and hydration: de Dios, Pearson, and Oldfield 1993,

Case 1995, Boyd and Skrynnikov 2002, Christensen, Sauer, and Jensen 2011, Baker 2001, Hass 2008, Kukic 2013, Nistor 2006, Caldeweyher 2019, Stone 2005, Kjaer 2011, Parker 2006, Larsen 2015, Grimme 2010, Wagner and Schreiner 2015, Nakasako 2004, Halle 2004, and Martin and Matyushov 2015. Facelli 2011 had Crossref metadata but no Europe PMC abstract text in this pass.

The older canon is treated as named lineage, as in the companion chapters: Ramsey 1950, Pople 1956, McConnell 1957, Johnson and Bovey 1958, Haigh and Mallion 1979, Buckingham 1960, Kabsch and Sander 1983, London 1937, and Kromhout and Linder 1969 have DOI metadata support, while Europe PMC abstract records were absent or incomplete in this pass. Their role here is to anchor the historical method names; the modern cited papers carry the accessible abstract-confirmed support around the code claims.

References

- [1] Ramsey, N. F. (1950). Magnetic shielding of nuclei in molecules. *Physical Review*, 78, 699–703. DOI: 10.1103/PhysRev.78.699.
- [2] Facelli, J. C. (2011). Chemical shift tensors: theory and application to molecular structural problems. *Progress in Nuclear Magnetic Resonance Spectroscopy*, 58, 176–201. DOI: 10.1016/j.pnmrs.2010.10.003.
- [3] de Dios, A. C., Pearson, J. G., and Oldfield, E. (1993). Secondary and tertiary structural effects on protein NMR chemical shifts: an ab initio approach. *Science*, 260, 1491–1496. DOI: 10.1126/science.8502992.
- [4] Pople, J. A. (1956). Proton magnetic resonance of hydrocarbons. *The Journal of Chemical Physics*, 24, 1111. DOI: 10.1063/1.1742701.
- [5] McConnell, H. M. (1957). Theory of nuclear magnetic shielding in molecules. I. Long-range dipolar shielding of protons. *The Journal of Chemical Physics*, 27, 226–229. DOI: 10.1063/1.1743676.
- [6] Johnson, C. E., and Bovey, F. A. (1958). Calculation of nuclear magnetic resonance spectra of aromatic hydrocarbons. *The Journal of Chemical Physics*, 29, 1012–1014. DOI: 10.1063/1.1744645.
- [7] Haigh, C. W., and Mallion, R. B. (1979). Ring current theories in nuclear magnetic resonance. *Progress in Nuclear Magnetic Resonance Spectroscopy*, 13, 303–344. DOI: 10.1016/0079-6565(79)80010-2.
- [8] Case, D. A. (1995). Calibration of ring-current effects in proteins and nucleic acids. *Journal of Biomolecular NMR*, 6, 341–346. DOI: 10.1007/BF00197633.
- [9] Boyd, J., and Skrynnikov, N. R. (2002). Calculations of the contribution of ring currents to the chemical shielding anisotropy. *Journal of the American Chemical Society*, 124, 1832–1833. DOI: 10.1021/ja016476f.
- [10] Christensen, A. S., Sauer, S. P. A., and Jensen, J. H. (2011). Definitive benchmark study of ring current effects on amide proton chemical shifts. *Journal of Chemical Theory and Computation*, 7, 2078–2084. DOI: 10.1021/ct2002607.
- [11] Buckingham, A. D. (1960). Chemical shifts in the nuclear magnetic resonance spectra of molecules containing polar groups. *Canadian Journal of Chemistry*, 38, 300–307. DOI: 10.1139/v60-040.
- [12] Baker, N. A., Sept, D., Joseph, S., Holst, M. J., and McCammon, J. A. (2001). Electrostatics of nanosystems: application to microtubules and the ribosome. *Proceedings of the National Academy of Sciences*, 98, 10037–10041. DOI: 10.1073/pnas.181342398.

- [13] Hass, M. A. S., Jensen, M. R., and Led, J. J. (2008). Probing electric fields in proteins in solution by NMR spectroscopy. *Proteins: Structure, Function, and Bioinformatics*, 72, 333–343. DOI: 10.1002/prot.21929.
- [14] Kukic, P., Farrell, D., McIntosh, L. P., Garcia-Moreno E., B., Jensen, K. S., Toleikis, Z., Teilum, K., and Nielsen, J. E. (2013). Protein dielectric constants determined from NMR chemical shift perturbations. *Journal of the American Chemical Society*, 135, 16968–16976. DOI: 10.1021/ja406995j.
- [15] Nistor, R. A., Polihronov, J. G., Muser, M. H., and Mosey, N. J. (2006). A generalization of the charge equilibration method for nonmetallic materials. *The Journal of Chemical Physics*, 125, 094108. DOI: 10.1063/1.2346671.
- [16] Caldeweyher, E., Ehlert, S., Hansen, A., Neugebauer, H., Spicher, S., Bannwarth, C., and Grimme, S. (2019). A generally applicable atomic-charge dependent London dispersion correction. *The Journal of Chemical Physics*, 150, 154122. DOI: 10.1063/1.5090222.
- [17] Stone, A. J. (2005). Distributed multipole analysis: stability for large basis sets. *Journal of Chemical Theory and Computation*, 1, 1128–1132. DOI: 10.1021/ct050190+.
- [18] Kjaer, H., Sauer, S. P. A., and Kongsted, J. (2011). Benchmarking the multipole shielding polarizability/reaction field approach to solvation against QM/MM: applications to the shielding constants of N-methylacetamide. *The Journal of Chemical Physics*, 134, 044514. DOI: 10.1063/1.3546033.
- [19] Kabsch, W., and Sander, C. (1983). Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22, 2577–2637. DOI: 10.1002/bip.360221211.
- [20] Parker, L. L., Houk, A. R., and Jensen, J. H. (2006). Cooperative hydrogen bonding effects are key determinants of backbone amide proton chemical shifts in proteins. *Journal of the American Chemical Society*, 128, 9863–9872. DOI: 10.1021/ja0617901.
- [21] Larsen, A. S., Bratholm, L. A., Christensen, A. S., Channir, M., and Jensen, J. H. (2015). ProCS15: a DFT-based chemical shift predictor for backbone and C beta atoms in proteins. *PeerJ*, 3, e1344. DOI: 10.7717/peerj.1344.
- [22] London, F. (1937). The general theory of molecular forces. *Transactions of the Faraday Society*, 33, 8–26. DOI: 10.1039/tf937330008b.
- [23] Grimme, S., Antony, J., Ehrlich, S., and Krieg, H. (2010). A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *The Journal of Chemical Physics*, 132, 154104. DOI: 10.1063/1.3382344.
- [24] Wagner, J. P., and Schreiner, P. R. (2015). London dispersion in molecular chemistry: reconsidering steric effects. *Angewandte Chemie International Edition*, 54, 12274–12296. DOI: 10.1002/anie.201503476.
- [25] Kromhout, R. A., and Linder, B. (1969). The effect of dispersion interaction on nuclear magnetic shielding. *Journal of Magnetic Resonance*, 1, 450–463. DOI: 10.1016/0022-2364(69)90081-X.
- [26] Nakasako, M. (2004). Water-protein interactions from high-resolution protein crystallography. *Philosophical Transactions of the Royal Society B*, 359, 1191–1206. DOI: 10.1098/rstb.2004.1498.
- [27] Halle, B. (2004). Protein hydration dynamics in solution: a critical survey. *Philosophical Transactions of the Royal Society B*, 359, 1207–1224. DOI: 10.1098/rstb.2004.1499.
- [28] Martin, D. R., and Matyushov, D. V. (2015). Dipolar nanodomains in protein hydration shells. *The Journal of Physical Chemistry Letters*, 6, 407–412. DOI: 10.1021/jz5025433.